

SERVER MANAGER — FEATURE INSTALLATION

Troubleshooting Guide & Standard Procedures
IT Department | Windows Server Administration

SECTION 1 — Pre-Installation Checks

Before installing any feature via Server Manager, verify the following:

#	Check Item	How to Verify
1	Server is fully updated	Run Windows Update — pending updates can block feature installation
2	Logged in as Administrator	Must be local or domain Administrator — not standard user
3	Sufficient disk space	Minimum 1-2GB free on C:\ — features extract files during install
4	Internet or WSUS access	Some features need source files from Windows Update or WSUS
5	No pending restart	Check Server Manager dashboard — pending restart blocks installs
6	Server Manager is up to date	Close and reopen Server Manager before starting

WARNING: Always take a snapshot or checkpoint (if VM) before installing roles/features on a production server.

SECTION 2 — Standard Feature Installation (GUI)

Step	Action	Details
1	Open Server Manager	Start → Server Manager (run as Administrator)
2	Click 'Add Roles and Features'	From Dashboard or Manage menu
3	Installation Type	Select 'Role-based or feature-based installation'
4	Select Server	Choose the target server from the pool
5	Select Feature	Navigate and check the feature/role to install
6	Add Required Features	Click 'Add Features' if prompted for dependencies
7	Confirm Installation	Review summary — tick 'Restart automatically if required'
8	Monitor Progress	Wait for installation to complete — do NOT close the wizard

9	Verify Installation	Check Server Manager dashboard — feature should show as Installed
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SECTION 3 — Install via PowerShell (Recommended Alternative)

If Server Manager GUI fails, use PowerShell — more reliable and shows detailed error output.

Check available features:

```
Get-WindowsFeature | Where-Object {$_.InstallState -eq 'Available'} | Select Name,
DisplayName
```

Install a specific feature:

```
Install-WindowsFeature -Name <FeatureName> -IncludeManagementTools -Verbose
```

Install with all sub-features:

```
Install-WindowsFeature -Name <FeatureName> -IncludeAllSubFeature -IncludeManagementTools
```

Common feature names:

PowerShell Feature Name	Description
AD-Domain-Services	Active Directory Domain Services
DNS	DNS Server
DHCP	DHCP Server
Web-Server	IIS Web Server
File-Services	File and Storage Services
RSAT-AD-Tools	AD Remote Server Admin Tools
Telnet-Client	Telnet Client
NET-Framework-45-Features	.NET Framework 4.5

SECTION 4 — Troubleshooting Common Errors

Use this table when feature installation fails or gets stuck.

Error / Symptom	Likely Cause	Fix / Action
Installation fails with error 0x800F0922	Cannot reach Windows Update or WSUS	Check internet/WSUS connectivity. Use -Source flag pointing to SxS folder.
'The source files could not be found'	Missing Windows installation source	Mount Windows ISO and run: Install-WindowsFeature -Source D:\sources\sxs
Greyed out / cannot select feature	Conflicting role already installed or pending restart	Check for pending restart. Reboot server first.
Installation stuck at 'Installing' forever	Background service hung or network issue	Wait 10 mins. If no progress, cancel and check CBS.log for errors.
'Access Denied' error	Not running as Administrator	Close Server Manager. Reopen as Administrator (right-click → Run as Admin).
Feature shows 'Failed' in dashboard	Incomplete install or dependency missing	Run: Get-WindowsFeature to check status. Reinstall via PowerShell with -Verbose.
Server restarts during install	Required restart was triggered	Normal behavior. Log back in — Server Manager will resume automatically.
'This operation is not supported'	Server Core or wrong server edition	Verify Windows Server edition. Some features require Desktop Experience.
CBS.log shows component store corruption	WinSXS component store corrupted	Run: DISM /Online /Cleanup-Image /RestoreHealth then retry install.
Dependency feature fails to install	Required dependency not met	Install dependencies first manually, then retry the main feature.

SECTION 5 — Component Store Repair (DISM + SFC)

Run these commands in order when source file errors or component corruption is suspected.

Step 1 — Check component store health:

```
DISM /Online /Cleanup-Image /CheckHealth
```

Step 2 — Scan for corruption:

```
DISM /Online /Cleanup-Image /ScanHealth
```

Step 3 — Restore health (requires internet or ISO source):

```
DISM /Online /Cleanup-Image /RestoreHealth
```

Step 4 — Run SFC after DISM completes:

```
sfc /scannow
```

Step 5 — If no internet, use Windows ISO as source:

```
DISM /Online /Cleanup-Image /RestoreHealth /Source:D:\sources\sxs /LimitAccess
```

WARNING: Always run PowerShell or CMD as Administrator when executing DISM and SFC commands.

SECTION 6 — Where to Find Log Files

Log File	Location	Purpose
CBS.log	C:\Windows\Logs\CBS\CBS.log	Component-Based Servicing — main install log
DISM.log	C:\Windows\Logs\DISM\dism.log	DISM operations and errors
ServerManager.log	C:\Windows\Logs\ServerManager.log	Server Manager GUI operations
WindowsUpdate.log	C:\Windows\WindowsUpdate.log	Windows Update activity
Event Viewer	eventvwr.msc → Windows Logs → Application/System	System-level install events

TIP: When escalating an issue, always attach CBS.log and DISM.log. These logs contain the exact error codes needed for diagnosis.